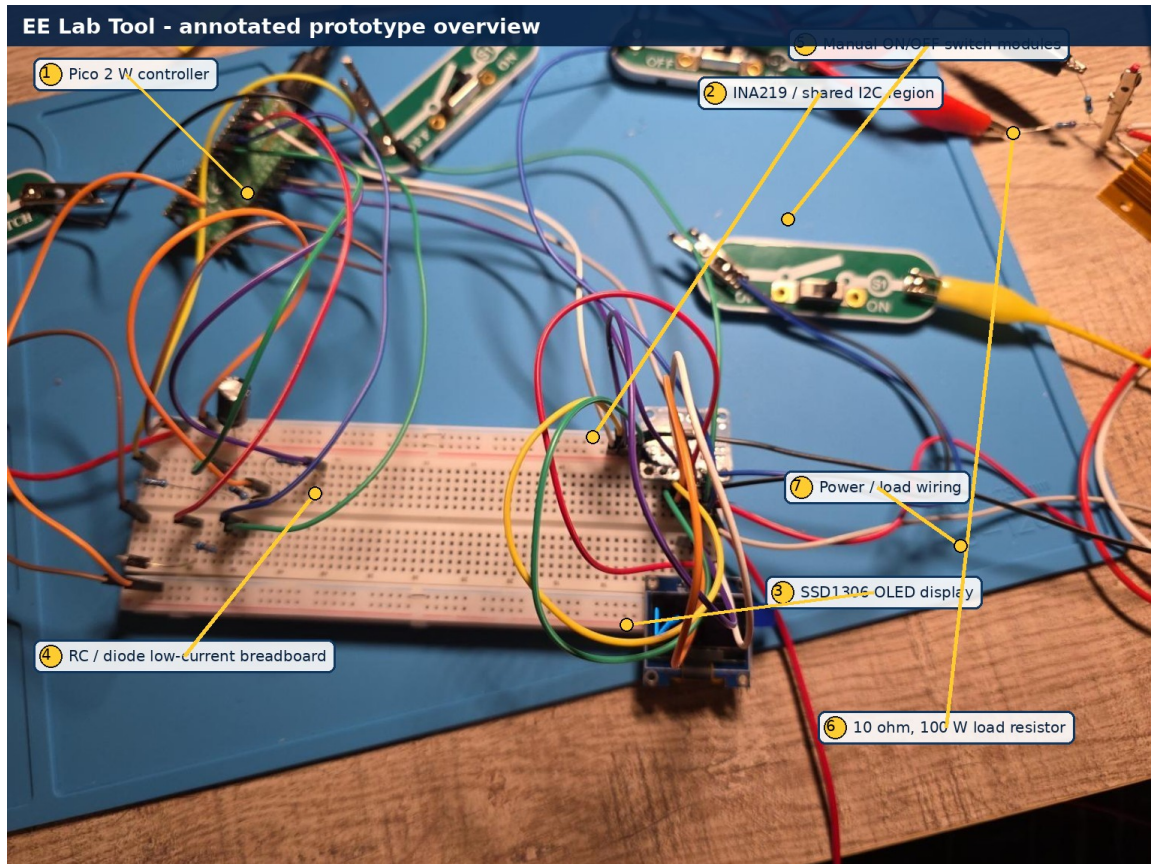


EE Summer Project Log

Raspberry Pi Pico 2 W EE Lab Tool

Final integrated documentation update - 27 August 2026

Power/Energy Monitor | RC Analyzer | Diode Analyzer | MCU Telemetry | Source Test | Fault Monitoring
| Logging/Export



Cover figure. Sanitized and annotated final prototype overview.

Repository target: Mini34/pico-2w-power-transient-analyzer

Executive summary

The Pico 2 W EE Lab Tool is a self-contained embedded instrumentation platform that grew from a DC power-monitoring proof of concept into a five-mode laboratory instrument with a Wi-Fi dashboard. The final working prototype combines calibrated power and energy measurement, RC transient characterization, diode testing, internal MCU-temperature telemetry, guided source characterization, threshold/fault monitoring, persistent experiment logging, and CSV/JSON export.

The project emphasizes measurement integrity. Defective INA219 modules, contradictory readings, ground-reference error, jumper/contact resistance, calibration outliers, RC wiring interactions, diode excitation-method errors, and an invalid negative source-resistance fit were investigated rather than hidden. The final documentation therefore records both successful results and the reasoning used to accept or reject measurements.

The dashboard and all software-side roadmap features were verified by the user. Automated electronic load stepping and a hardware cutoff stage remain future hardware work. The current Source Test is guided/manual, and its negative-resistance example is retained only as an invalid-result case that the repository should flag explicitly.

Current capability status

Capability	Status	Technical scope
Power / Energy Monitor	Complete	Live V/I/P, mAh, Wh, min/peak/average statistics, OLED pages, dashboard, session logging.
RC Analyzer	Complete	Automated charge/discharge, live OLED curves, tau and capacitance, dashboard traces, logging.
Diode Analyzer	Complete with method boundary	Steady-DC single test is quantitative; PWM I-V sweep is display-only.
Wi-Fi dashboard	Complete	Live cards, charts, mode/state, history, exports, system status.
MCU temperature telemetry	Complete	Internal die telemetry; not ambient; intentionally idle in STARTUP/MENU.
Experiment logging/export	Complete	Power, RC, diode, source, and fault records; CSV/JSON downloads.
Fault/threshold monitoring	Complete as software monitoring	No hardware cutoff/protection stage.
Guided Source Test	Implemented; validation open	Manual point capture and regression work; negative-Rs results are invalid and flagged.

Verified architecture and pin map

EE LAB TOOL - VERIFIED SYSTEM ARCHITECTURE

Raspberry Pi Pico 2 W | MicroPython | SoftI2C 50 kHz | Wi-Fi dashboard

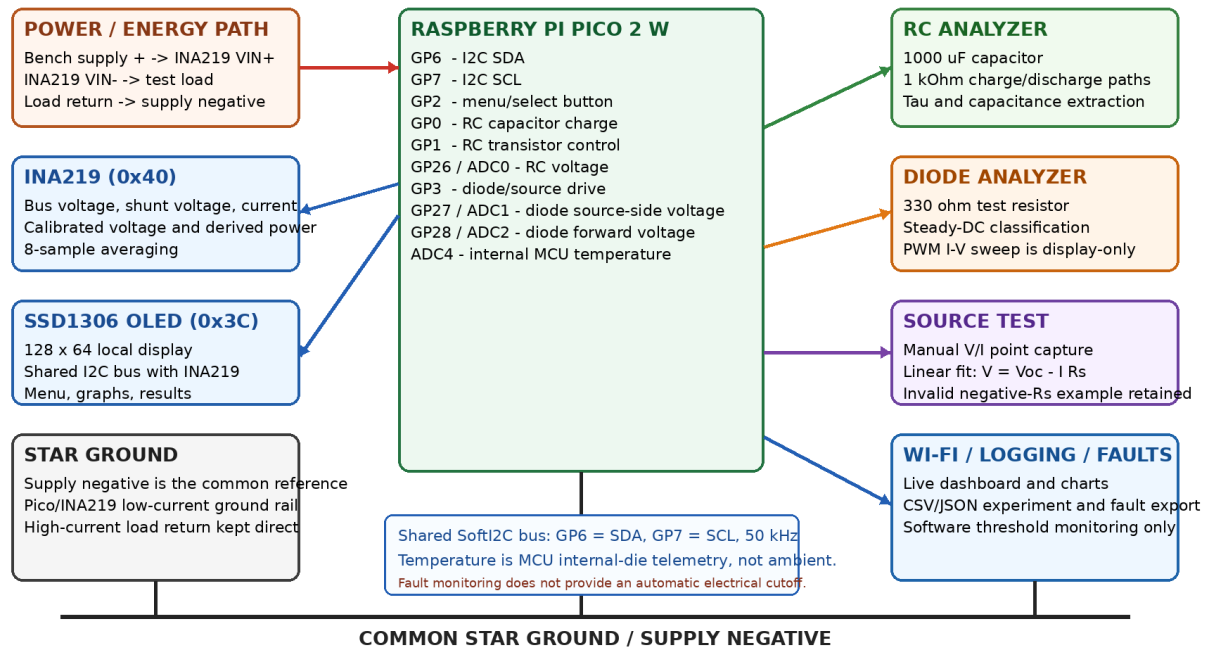


Figure 1. Verified system architecture. Exact firmware uses SoftI2C at 50 kHz on GP6/GP7.

Pico resource	Function
GP0	RC capacitor charge
GP1	RC transistor control / discharge path
GP2	menu/select button, active low
GP3	diode test source / PWM sweep drive
GP6	I2C SDA
GP7	I2C SCL
GP26_ADC0	RC capacitor voltage
GP27_ADC1	diode source-side voltage
GP28_ADC2	diode forward voltage
ADC4	RP2350 internal temperature sensor

Calibration and measurement integrity

- Final voltage equation: $V_{cal} = 0.98896 * V_{raw} - 0.03531$ V. Fit quality R-squared approximately 0.999975.
- The accepted calibration residuals were approximately plus/minus 3 mV, near the practical resolution of the INA219 bus-voltage path.
- Direct R100 shunt measurement gave 30.1-30.2 mV, corresponding to approximately 0.301-0.302 A; the INA219 reported approximately 0.300 A. No current correction factor was justified.
- Star grounding reduced ground-reference error from approximately 86 mV to approximately 0.2 mV.

- Three parallel temporary positive jumpers reduced positive-path drop from approximately 0.39 V to approximately 0.076 V. Contact and wiring losses remain real components of the prototype.
- Alligator clips were experimentally worse than the curled/stripped-wire load connection in the earlier high-current tests, reinforcing that visible connector size is not a guarantee of low contact resistance.

1. Hardware bring-up and soldering

Soldered Pico and INA219 headers, recovered from oxidized-tip and breadboard/contact issues, and established stable power and ground rails.

Evidence

- Pico 3V3 measured approximately 3.22-3.26 V
- Shared I2C bus ultimately detected both devices

Decision / next step: Continue with GP6/GP7 shared I2C and preserve one-step-at-a-time physical verification.

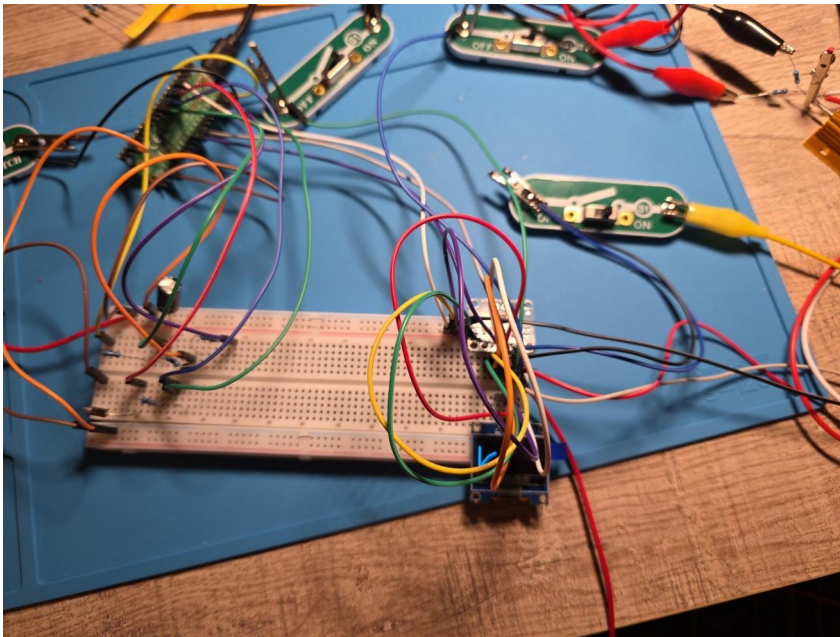


Figure 1. Overall prototype overview showing the complete Pico 2 W EE Lab Tool on the work mat, including the breadboard, external switch modules, OLED, and surrounding test wiring.

2. I2C and module diagnosis

Separated communication faults from measurement faults. A scan returned [60, 64], proving OLED 0x3C and INA219 0x40 communication.

Evidence

- First INA219 produced implausible bus voltage despite valid current
- Third tested module produced realistic bus-voltage data

Decision / next step: Reject defective/noncompliant modules rather than calibrating around hardware failure.

3. Power-monitor calibration and wiring integrity

Moved from a 500 ohm load to a 10 ohm, 100 W load, quantified real connection losses, implemented star grounding, and fitted the voltage calibration.

Evidence

- Ground-reference error reduced from approximately 86 mV to approximately 0.2 mV
- Positive-path drop reduced from approximately 0.39 V to approximately 0.076 V using three parallel temporary jumpers
- Voltage fit R^2 approximately 0.999975

Decision / next step: Treat parasitic resistance as a physical circuit element; do not calibrate it away.

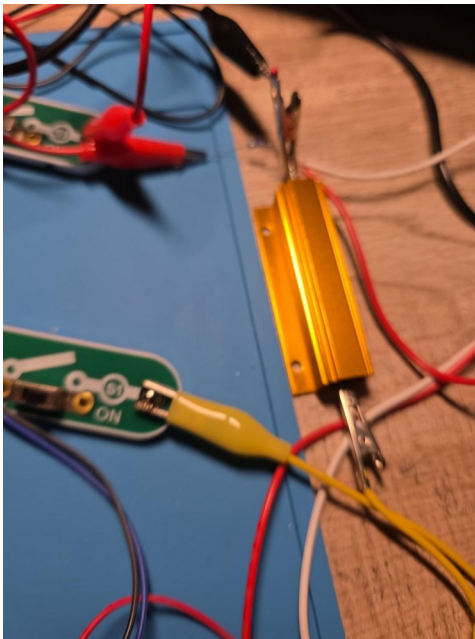


Figure 8. 10 ohm, 100 W aluminum power resistor used as the principal high-current load.

4. Current-channel verification

Rejected an inconsistent multimeter 10 A range result and measured directly across the INA219 R100 shunt.

Evidence

- 30.1-30.2 mV across 0.1 ohm implies approximately 0.301-0.302 A
- INA219 simultaneously reported approximately 0.300 A

Decision / next step: No software current scale factor is justified.

5. Power and energy analyzer

Extended the Power Monitor with charge and energy integration, minimum/peak statistics, running averages, elapsed time, OLED pages, dashboard state, and end-of-session logging.

Evidence

- Final dashboard example: 2.996 V, 0.469 A, 1.404 W
- 9.70 mAh and 0.0292 Wh over a 103 s example session

Decision / next step: Power/Energy Monitor accepted as a completed subsystem.

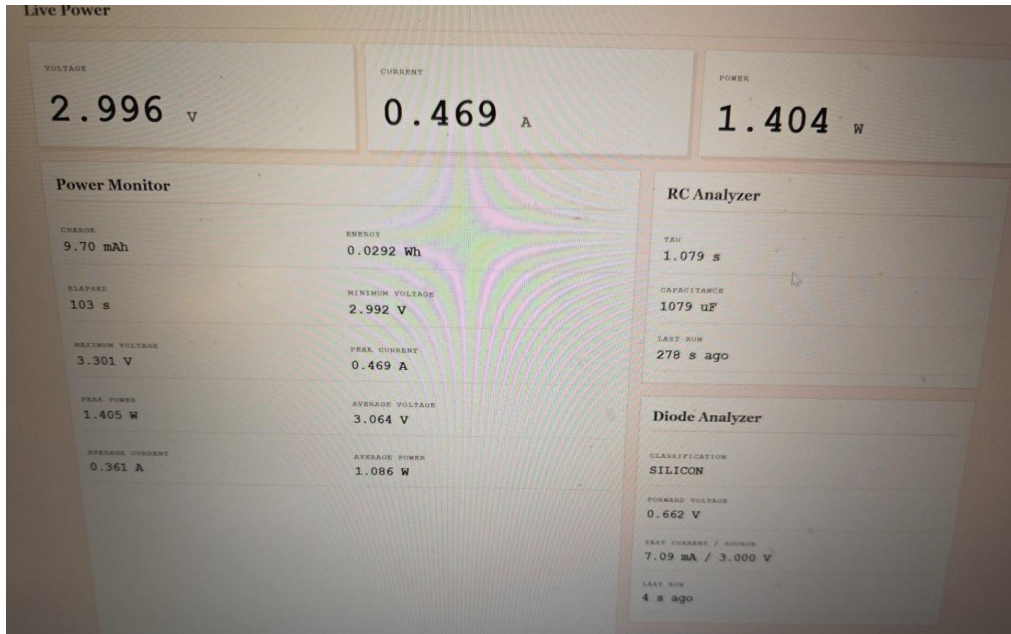


Figure 21. Sanitized dashboard view showing live voltage/current/power, Power Monitor session statistics, RC Analyzer results, and the silicon-diode result.

6. RC analyzer

Built a 2N2222-controlled RC charge/discharge network using a 1000 uF capacitor and 1 kOhm paths. Added live OLED curves, 1/e crossing interpolation, capacitance calculation, dashboard traces, and result logging.

Evidence

- Measured tau approximately 1.076-1.079 s
- Calculated capacitance approximately 1076-1079 uF
- Charge curve agreed with the 63.2% first-order checkpoint

Decision / next step: RC Analyzer accepted as a completed subsystem.

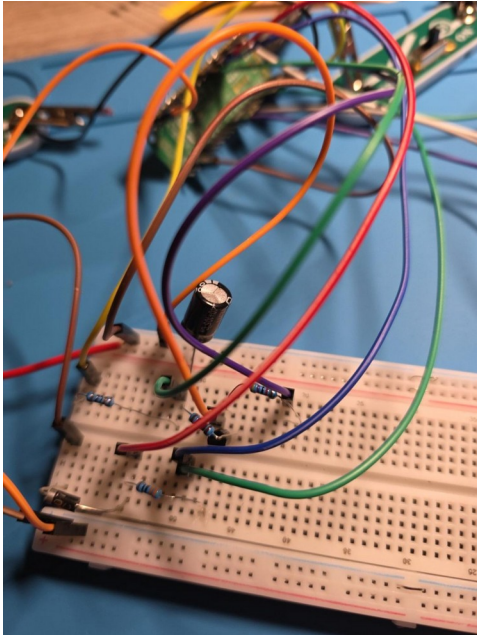


Figure 3. RC analyzer section close-up showing the 1000 μ F electrolytic capacitor, resistors, transistor, and ADC/control wiring.

7. Diode analyzer

Added settled single-point diode measurement and a PWM-driven I-V visualization. Corrected an early method error by restricting classification to the settled DC result.

Evidence

- 1N4007 silicon example: V_f approximately 0.662 V at 7.09 mA from a 3.000 V source
- 1N5819 Schottky example: V_f approximately 0.248 V at 8.20 mA from a 2.954 V source
- I-V sweep displays 15 points but is not used for quantitative classification

Decision / next step: Single-test classification is quantitative; PWM sweep is visualization-only until an analog source is validated.

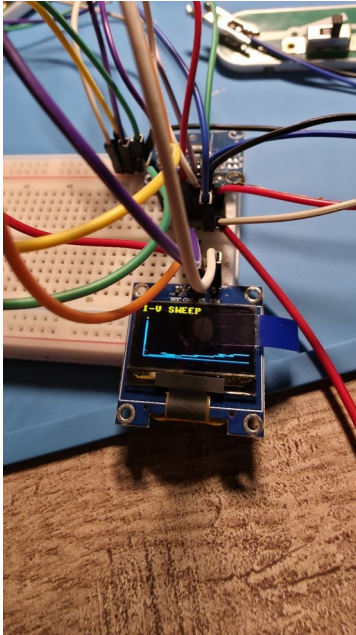


Figure 15. OLED I-V sweep visualization from the Diode Analyzer.

8. Unified instrument UI

Combined Power, RC, Diode, MCU Temperature, and Source Test modes into one startup/menu workflow controlled by one active-low button.

Evidence

- Short press navigates/pages; long press selects/returns
- OLED screens and photographed menu/result pages verified

Decision / next step: Routine operation no longer depends on separate Thonny scripts.

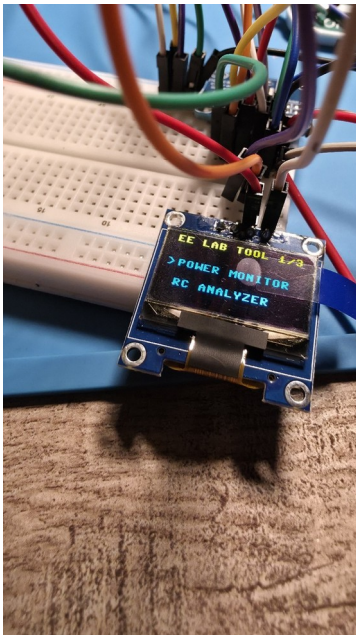


Figure 11. OLED main-menu page showing the unified EE LAB TOOL interface and analyzer selection.

9. Wi-Fi dashboard

Integrated Wi-Fi connection management, shared instrument state, HTTP serving, live status cards, power history, RC traces, diode results, source-test state, MCU telemetry, and logs.

Evidence

- Dashboard connected and served on the local network
- User verified all live panels and charts

Decision / next step: Networking/dashboard support treated as a completed subsystem.

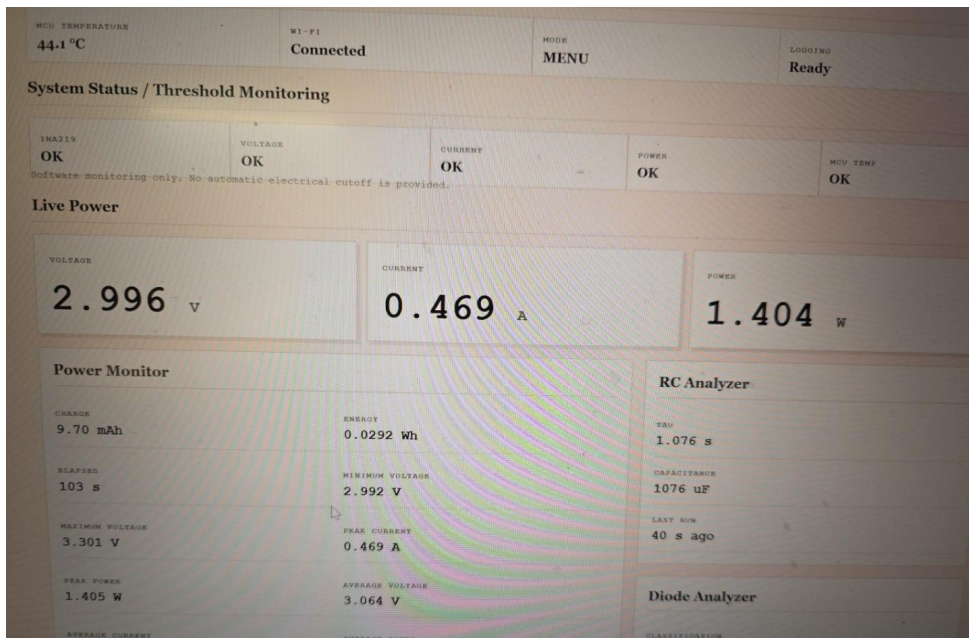


Figure 10. Sanitized dashboard overview showing MCU temperature, Wi-Fi status, threshold monitoring, live power, Power Monitor statistics, and RC Analyzer results.

10. MCU temperature telemetry

Read ADC4 with 16-sample averaging every five seconds when the instrument is in an active mode. Corrected TEMP_VREF to 3.24 V and exposed the reading to OLED/dashboard state.

Evidence

- Stable readings approximately 45.53-45.68 C
- ADC approximately 0.6739-0.6741 V, raw approximately 13630-13635

Decision / next step: Label as MCU/internal-die temperature, not ambient. Preserve the intentional pause in STARTUP and MENU.



Figure 24. Sanitized dashboard view showing RC traces, MCU internal-die temperature history, and experiment/fault export controls.

11. Experiment logging and export

Added persistent completed-test summaries for power sessions, RC tests, diode single tests, diode I-V sweeps, source tests, and faults. Added dashboard history plus CSV/JSON downloads.

Evidence

- Experiment table displays type, status, uptime, and summaries
- Separate experiment/fault CSV and JSON export controls verified

Decision / next step: Logging/export accepted as a completed software subsystem.

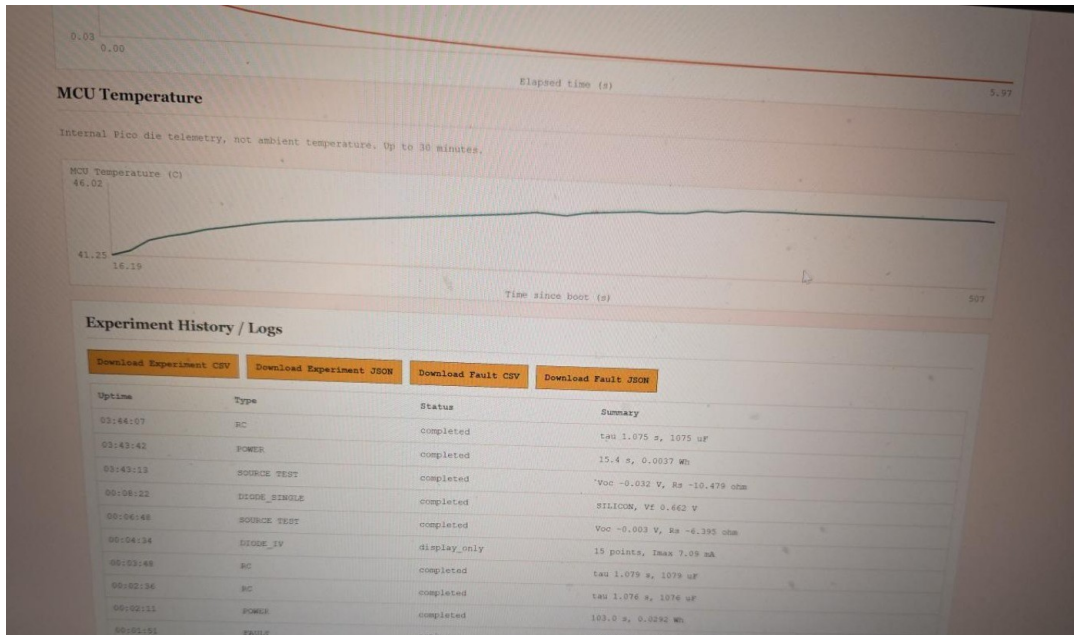


Figure 25. Sanitized dashboard view showing MCU temperature history and the experiment-history table with CSV/JSON export controls.

12. Guided Source Test

Added manual capture of multiple INA219 V/I operating points and a reusable linear-fit backend for Voc, source resistance, and R^2 .

Evidence

- Point acceptance and duplicate rejection photographed
- One three-point test returned Voc -0.003 V, Rs -6.395 ohm, R^2 1.000

Decision / next step: Retain mode as guided/manual. Flag negative resistance as invalid; add stronger capture workflow and fit validation before treating results as physical.

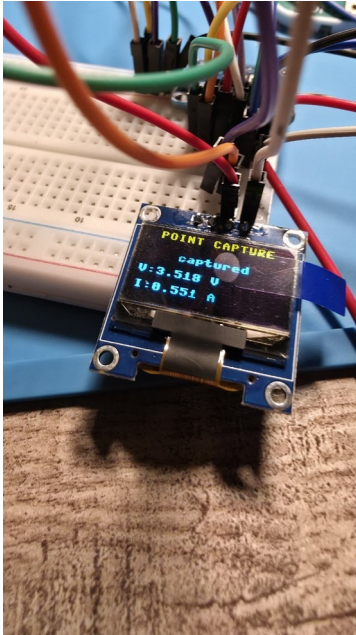


Figure 13. OLED Point Capture confirmation with the measured voltage and current for an accepted source-test point.

13. Fault and threshold monitoring

Added configurable software monitoring for undervoltage, overcurrent, excessive power, MCU temperature, and INA219 disconnect/restoration, with state transitions and event logging.

Evidence

- System status dashboard verified
- UNDERVOLTAGE_ACTIVE example appears in history

Decision / next step: State clearly that this is monitoring only; no hardware cutoff is presently installed.

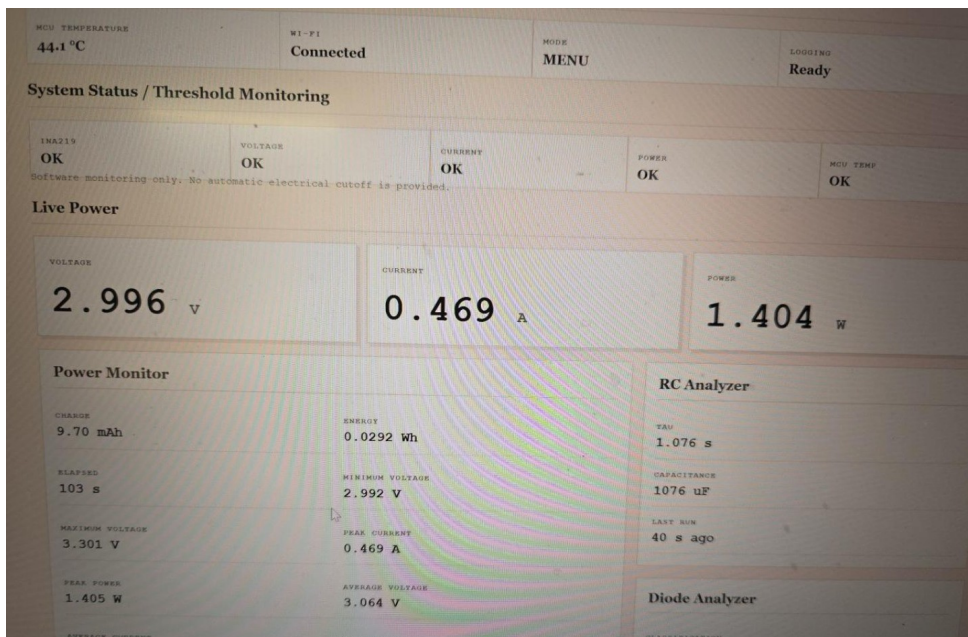


Figure 10. Sanitized dashboard overview showing MCU temperature, Wi-Fi status, threshold monitoring, live power, Power Monitor statistics, and RC Analyzer results.

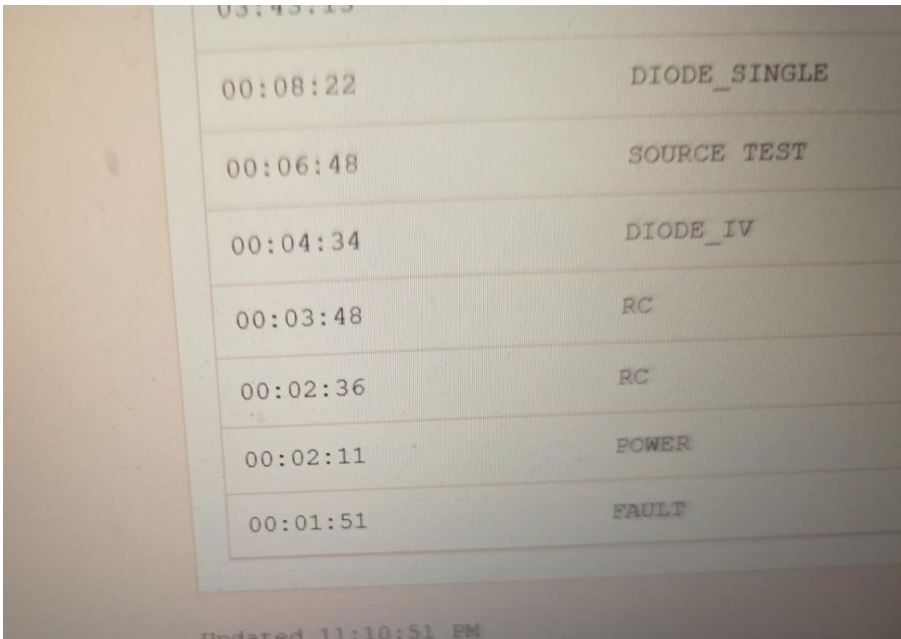
14. Final photographic/documentation capture

Captured the full hardware setup, close-ups of the controller, RC section, switches, load resistor, harness, OLED workflows, dashboard charts, telemetry, and logs. Sanitized screen photos to remove laptop branding, browser chrome, bookmarks, tabs, and local IP information.

Evidence

- 26 sanitized photographs in the final appendix
- Annotated overview, exact architecture diagram, and conceptual 3D visual prepared

Decision / next step: Use photos as engineering evidence with captions explaining what each image proves.



A photograph of a digital display showing a table of experiment history. The table has two columns: a timestamp column on the left and a description column on the right. The data rows are as follows:

Timestamp	Description
00:08:22	DIODE_SINGLE
00:06:48	SOURCE_TEST
00:04:34	DIODE_IV
00:03:48	RC
00:02:36	RC
00:02:11	POWER
00:01:51	FAULT

Below the table, an update timestamp is visible: Updated 11:10:51 PM.

Figure 26. Sanitized close-up of the experiment-history table and update timestamp.

Final measured examples

Subsystem	Photographed/verified example	Interpretation
Power / Energy	2.996 V, 0.469 A, 1.404 W; 9.70 mAh; 0.0292 Wh over 103 s	Working live measurement and session statistics.
RC Analyzer	Tau 1.076-1.079 s; capacitance 1076-1079 uF	Consistent with the nominal 1000 uF / 1 kOhm network and component tolerance.
Silicon diode	Vf about 0.662 V at 7.09 mA from a 3.000 V source	Correct steady-DC SILICON classification.
Schottky diode	1N5819 Vf about 0.248 V at 8.20	Validated Schottky comparison

	mA from a 2.954 V source	point.
MCU temperature	Approximately 45.53-45.68 C in the stable test; dashboard approximately 41.25-46.02 C	Internal die telemetry only; not ambient.
Source Test	Voc -0.003 V, Rs -6.395 ohm, R-squared 1.000, 3 points	Mathematically linear but physically invalid for the assumed source model; must be flagged.

Firmware and dashboard architecture

The supplied integrated main file coordinates hardware acquisition and modular services. The exact working copies of all imported modules must be exported from the active project workspace to GitHub. The current module set is:

- main.py
- instrument_state.py
- wifi_manager.py
- web_server.py
- experiment_logger.py
- fault_monitor.py
- source_characterizer.py
- ssd1306.py

Timing-sensitive RC and diode acquisitions suppress web/temperature service calls. Temperature sampling is intentionally disabled in STARTUP and MENU because continuous idle measurement was unwanted. Optional-service import and runtime failures are caught so that the core instrument can continue operating.

Known limitations and future hardware work

- Source Test can return a mathematically perfect but physically invalid negative source resistance when captured points or current direction do not represent the assumed $V = V_{oc} - I R_s$ model. Negative R_s must be rejected or clearly marked invalid.
- The diode PWM I-V sweep averages a pulsed waveform and is display-only. Steady-DC single-test data remains the classification source.
- MCU temperature is the RP2350 internal die temperature, not ambient temperature.
- Fault monitoring is software monitoring and logging only; it does not provide a hardware safety cutoff.
- High-current prototype wiring and clip/contact resistance remain measurable. Final construction should use robust low-resistance terminals/conductors.
- Automated electronic load stepping is not yet implemented. Current Source Test is guided/manual until MOSFET/load-bank hardware is added.
- Temperature sampling is intentionally deferred in STARTUP and MENU to avoid continuous background measurements when idle.

Future hardware work should proceed directly to a properly rated 3.3 V logic-level MOSFET switched-load bank if automated load stepping is desired. The small manual switches can remain as low-current circuit selectors, but their contact rating must be known before using them in the several-hundred-milliamp load path.

Final documentation and GitHub update package

This final package updates the prior repository snapshot without erasing its historical record. It includes a replacement README, a comprehensive Markdown log, a machine-readable JSON log, final DOCX/PDF documentation, a complete sanitized photo appendix, design assets, the current main.py, and explicit Codex instructions for completing the repository update with the remaining exact working firmware modules.

All current dashboard photos were cropped to remove laptop branding, browser tabs/bookmarks, the photographed local IP address, and unrelated user-interface content.

Appendix A - Complete sanitized photographic record

All 26 uploaded photographs are included below. Captions state what is visible and why the image matters. Full-resolution images are also provided separately in the photo appendix PDF and image directory.

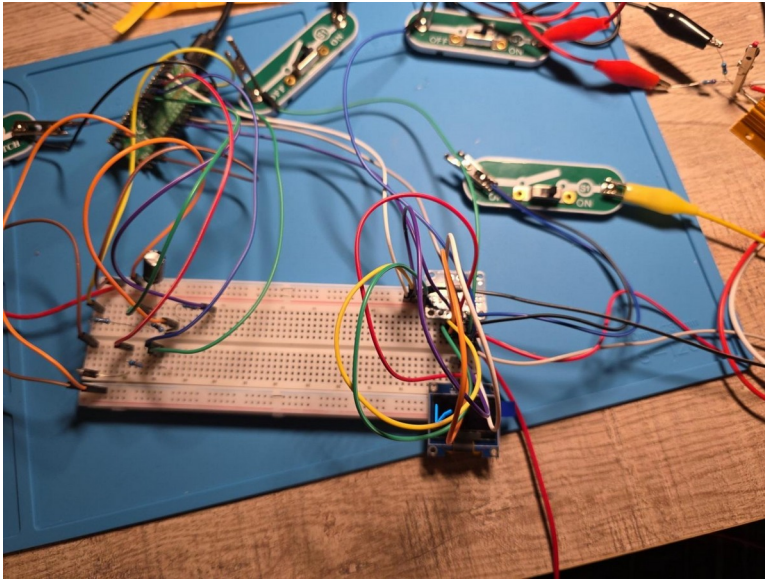


Figure 1. Overall prototype overview showing the complete Pico 2 WEE Lab Tool on the work mat, including the breadboard, external switch modules, OLED, and surrounding test wiring. Privacy action: none required.

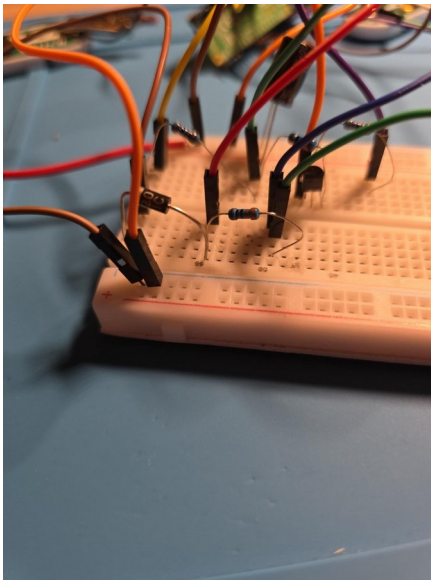


Figure 2. Close-up of the low-current breadboard test area, preserving component placement and jumper routing around the transistor/resistor network. Privacy action: none required.

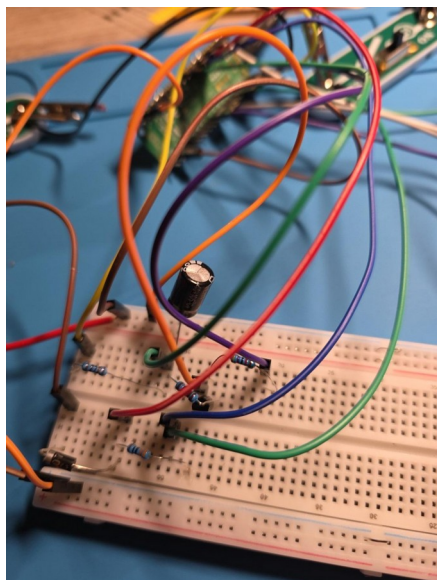


Figure 3. RC analyzer section close-up showing the 1000 μF electrolytic capacitor, resistors, transistor, and ADC/control wiring. Privacy action: none required.

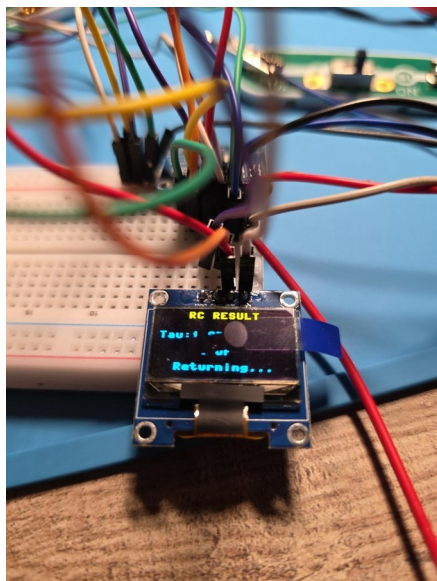


Figure 4. OLED RC Analyzer result screen documenting a successful measured time constant and capacitance result. Privacy action: none required.

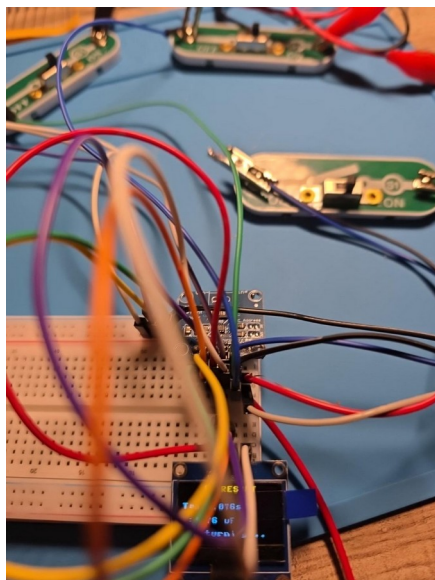


Figure 5. Top-side close-up of the Pico 2 W, INA219/OLED interconnections, and dense shared-bus/control wiring. Privacy action: none required.

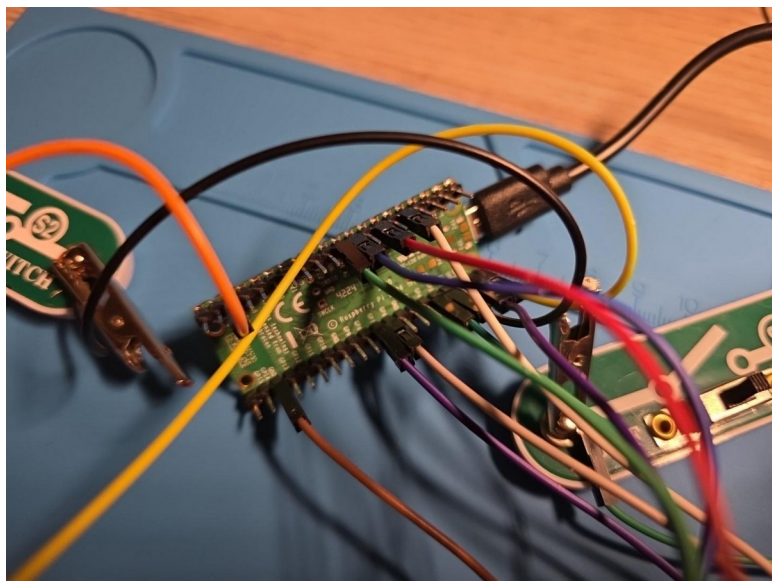


Figure 6. Raspberry Pi Pico 2 W controller close-up documenting the header connections used by the integrated instrument. Privacy action: none required.

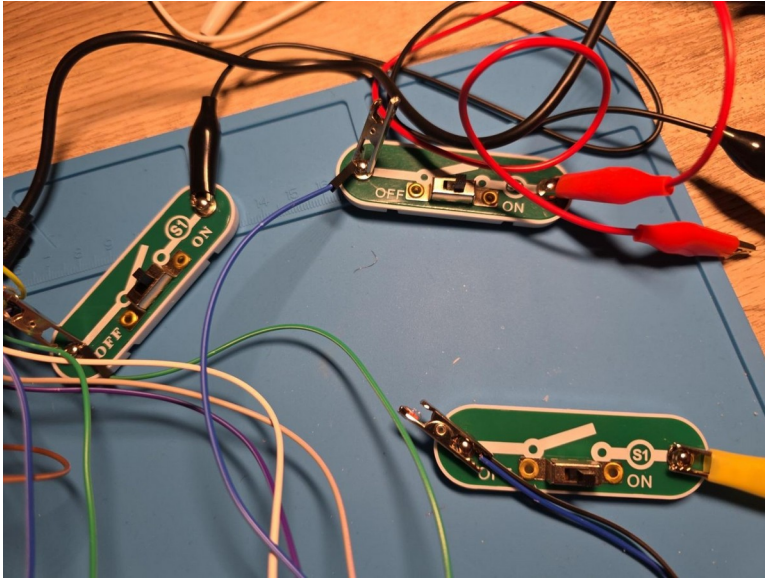


Figure 7. Manual ON/OFF switch modules and alligator-clip wiring used as physical circuit selectors/bypass elements during testing. Privacy action: none required.

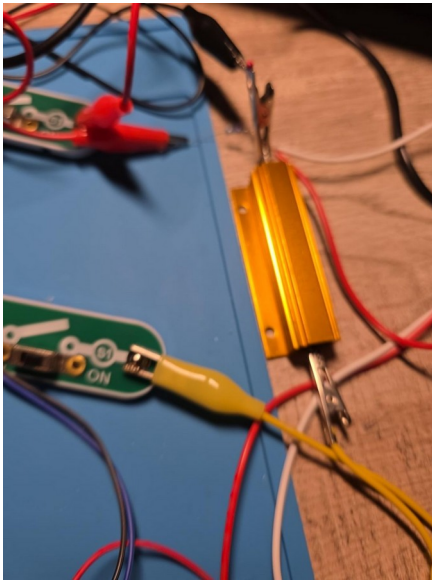


Figure 8. 10 ohm, 100 W aluminum power resistor used as the principal high-current load. Privacy action: none required.

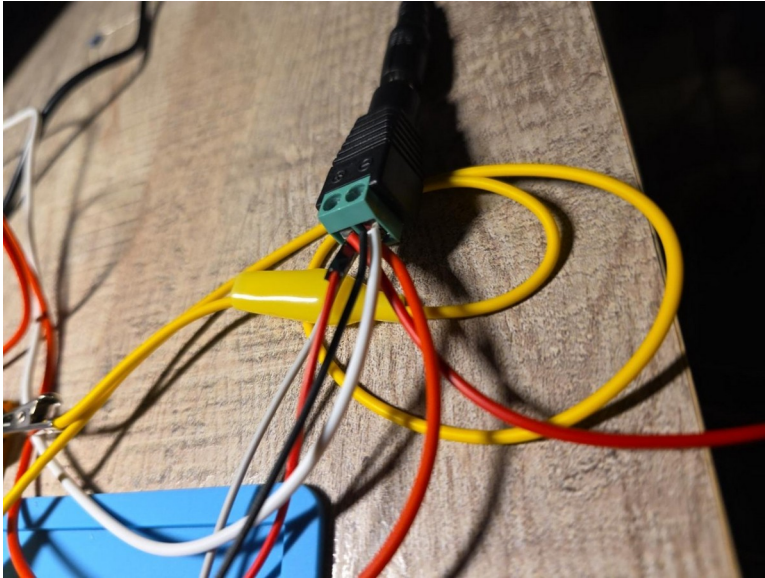


Figure 9. Screw-terminal power-harness breakout and multi-conductor wiring used for detachable source/load connections. Privacy action: none required.

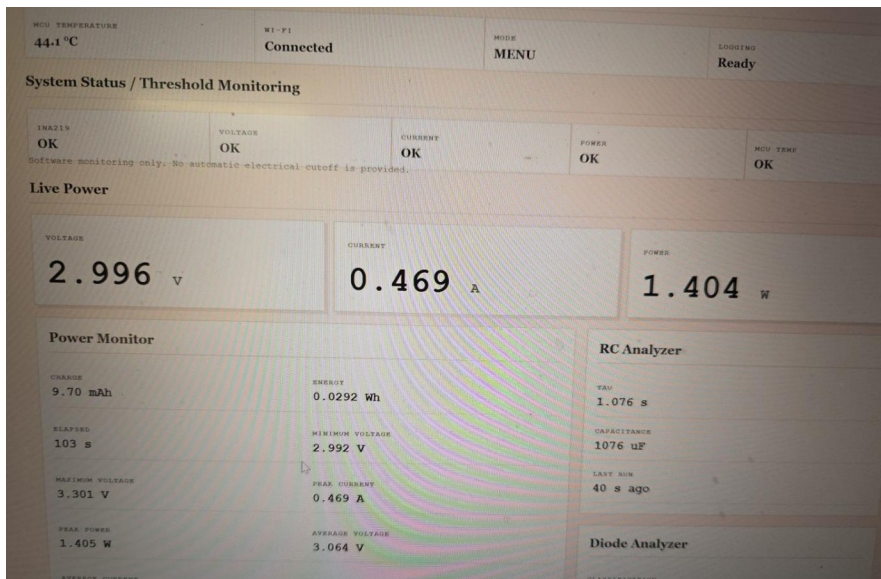


Figure 10. Sanitized dashboard overview showing MCU temperature, Wi-Fi status, threshold monitoring, live power, Power Monitor statistics, and RC Analyzer results. Privacy action: cropped to remove browser/laptop identifiers.

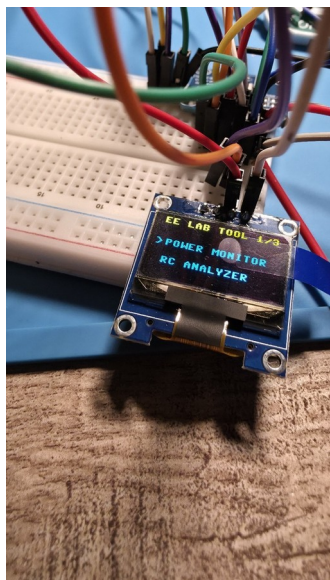


Figure 11. OLED main-menu page showing the unified EE LAB TOOL interface and analyzer selection. Privacy action: none required.

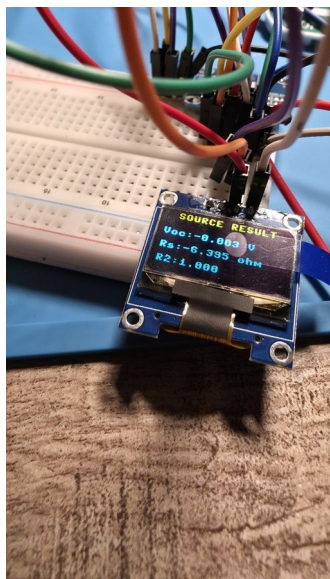
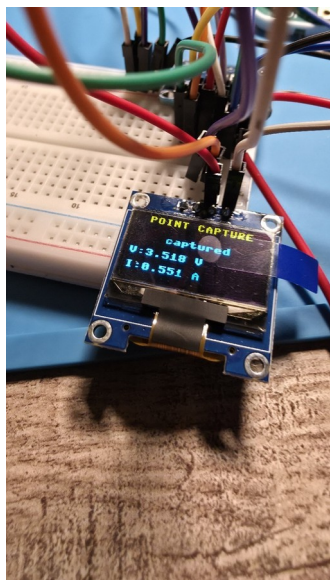


Figure 12. OLED Source Test result screen. The photographed negative source-resistance result is retained as a validation issue, not accepted as a physical source model. Privacy action: none required.



*Figure 13. OLED Point Capture confirmation with the measured voltage and current for an accepted source-test point.
Privacy action: none required.*

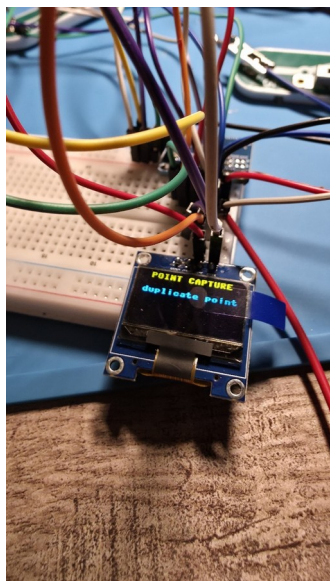


Figure 14. OLED duplicate-point rejection message demonstrating source-test data-quality screening. Privacy action: none required.

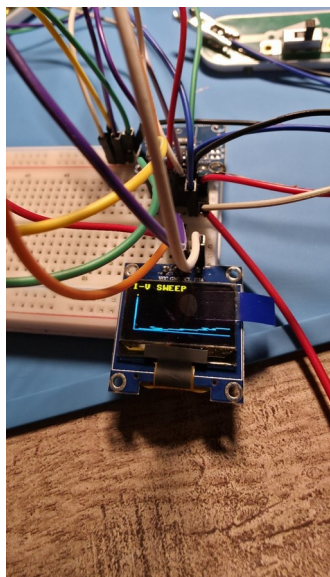


Figure 15. OLED I-V sweep visualization from the Diode Analyzer. Privacy action: none required.

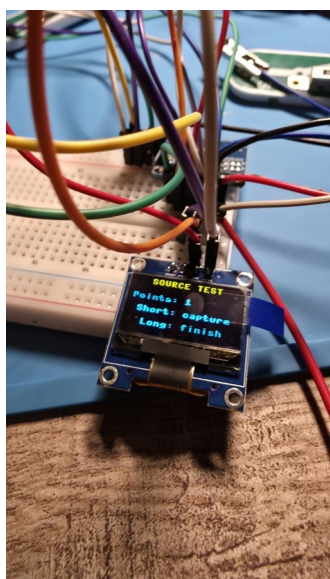


Figure 16. OLED Source Test collection screen showing the current point count and short/long press controls. Privacy action: none required.

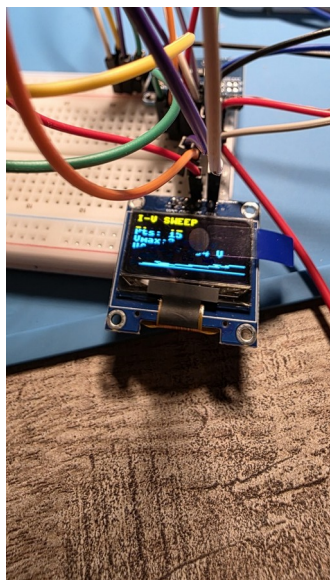


Figure 17. OLED I-V sweep numerical/statistical page documenting the captured sweep summary. Privacy action: none required.

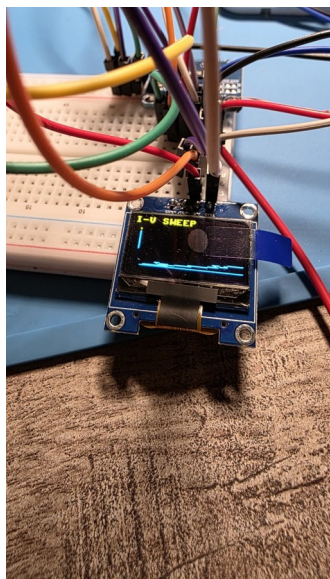


Figure 18. OLED I-V sweep graph, alternate captured frame. Privacy action: none required.

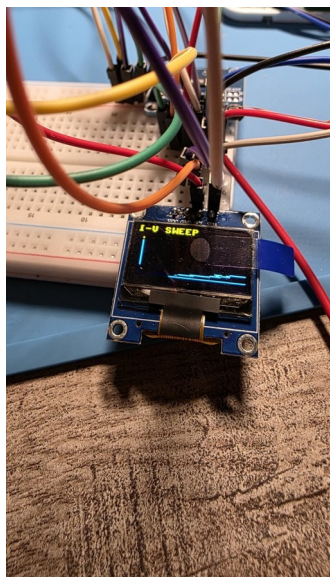


Figure 19. OLED I-V sweep graph, alternate captured frame preserving the displayed curve. Privacy action: none required.

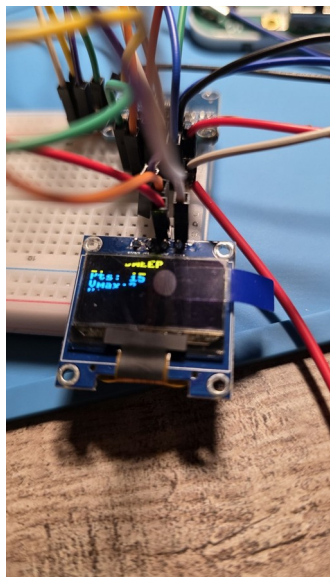


Figure 20. OLED I-V sweep statistics/graph frame documenting the embedded visualization workflow. Privacy action: none required.

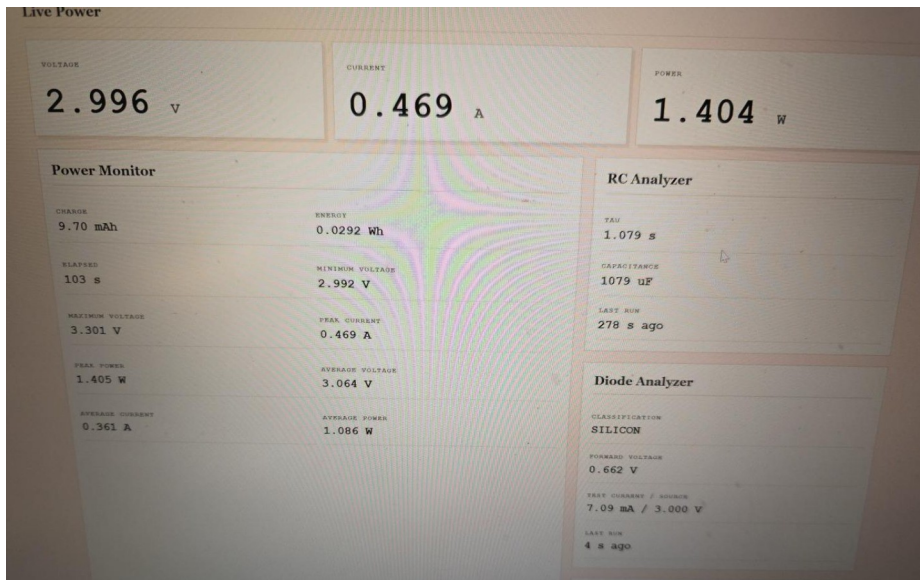


Figure 21. Sanitized dashboard view showing live voltage/current/power, Power Monitor session statistics, RC Analyzer results, and the silicon-diode result. Privacy action: cropped to remove browser/laptop identifiers.

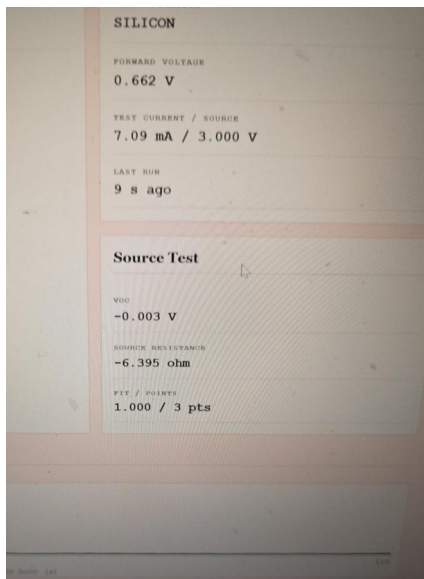


Figure 22. Sanitized dashboard close-up showing the silicon-diode result and the three-point Source Test fit that produced a negative resistance and therefore requires validation. Privacy action: cropped to remove browser/laptop identifiers.

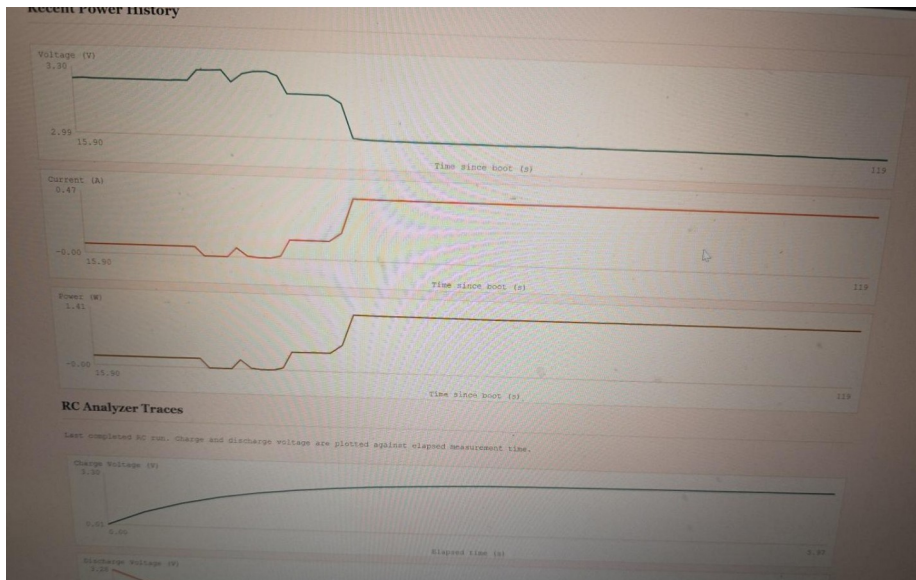


Figure 23. Sanitized dashboard charts showing recent voltage/current/power history and RC charge/discharge traces. Privacy action: cropped to remove browser/laptop identifiers.



Figure 24. Sanitized dashboard view showing RC traces, MCU internal-die temperature history, and experiment/fault export controls. Privacy action: cropped to remove browser/laptop identifiers.

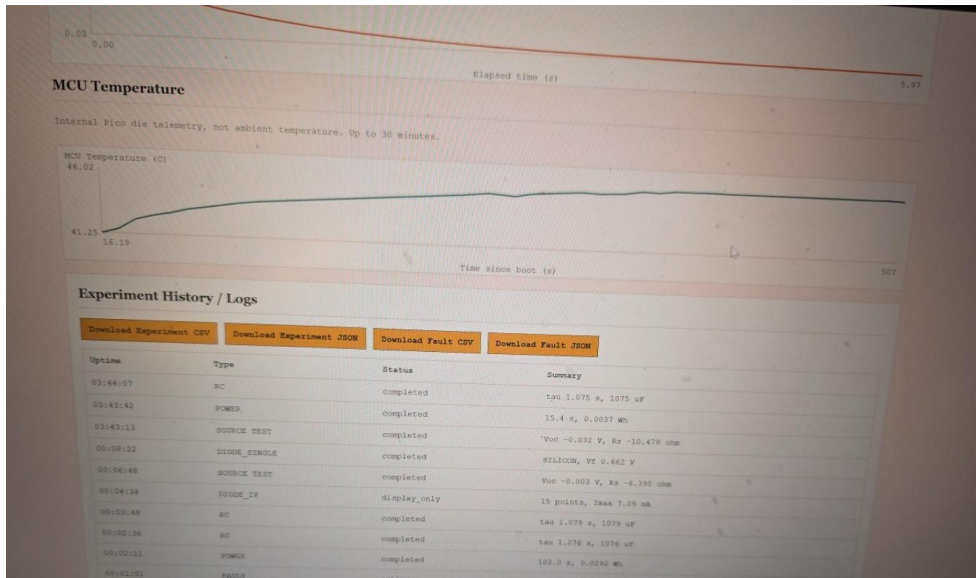


Figure 25. Sanitized dashboard view showing MCU temperature history and the experiment-history table with CSV/JSON export controls. Privacy action: cropped to remove browser/laptop identifiers.

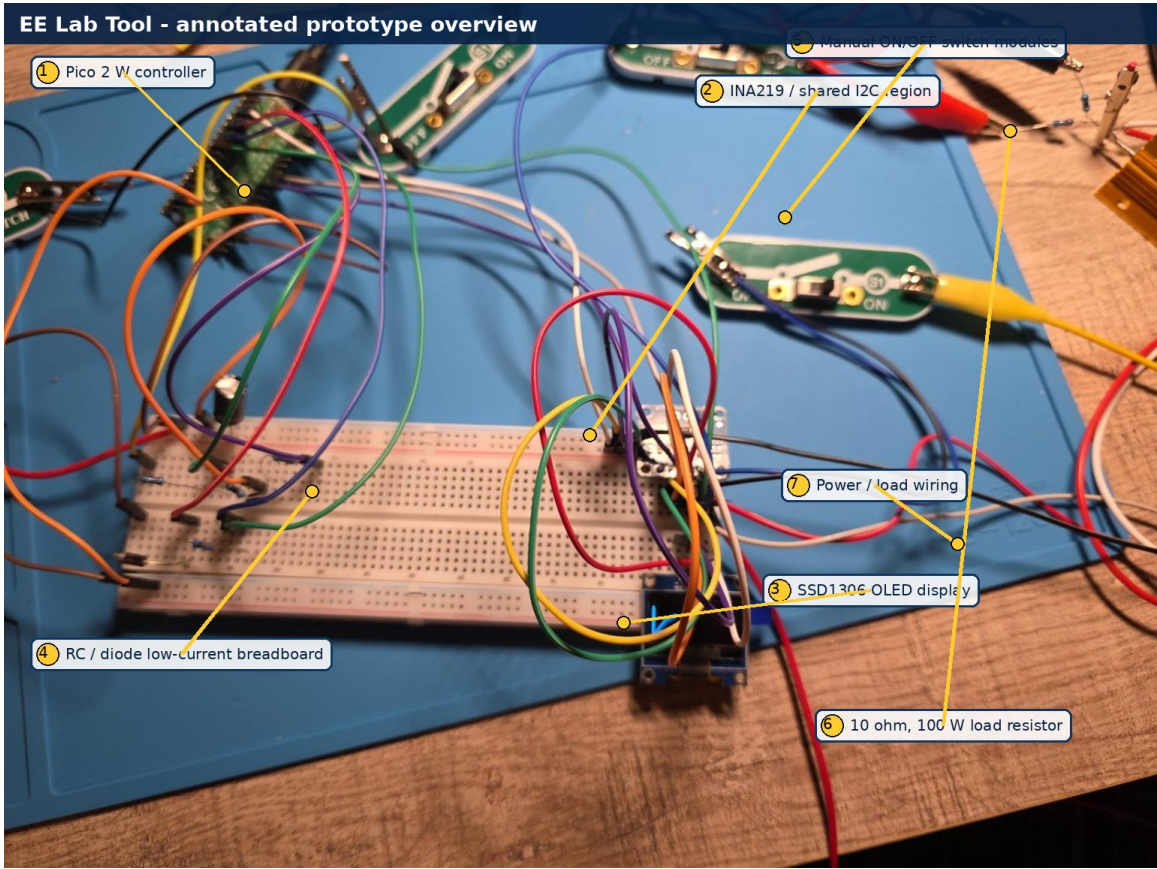
03:43:13

00:08:22	DIODE_SINGLE
00:06:48	SOURCE TEST
00:04:34	DIODE_IV
00:03:48	RC
00:02:36	RC
00:02:11	POWER
00:01:51	FAULT

Updated 11:10:51 PM

Figure 26. Sanitized close-up of the experiment-history table and update timestamp. Privacy action: cropped to remove browser/laptop identifiers.

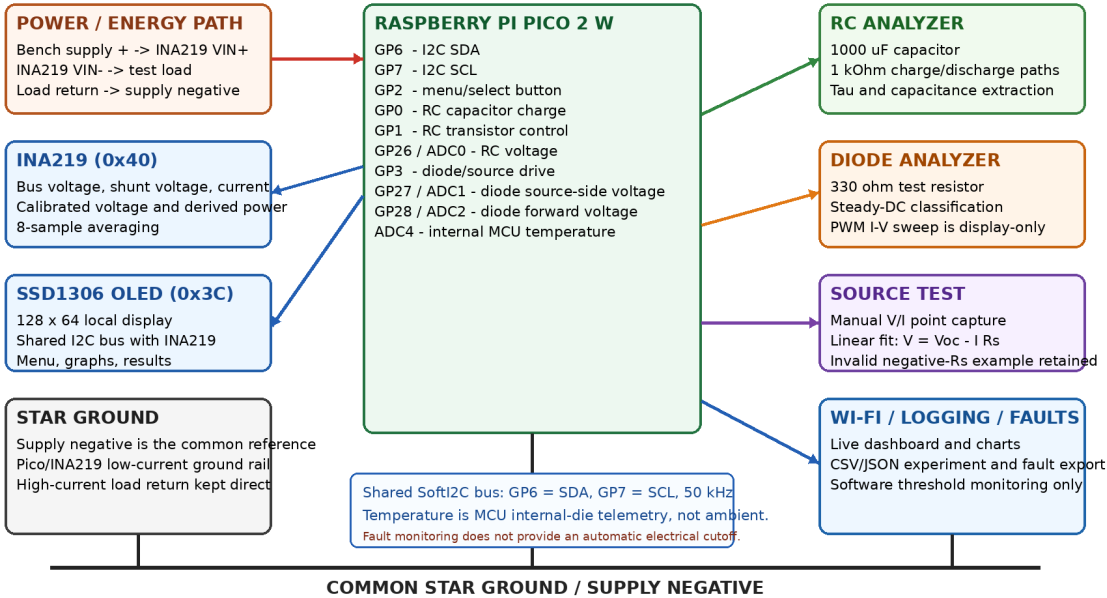
Appendix B - Digital design assets



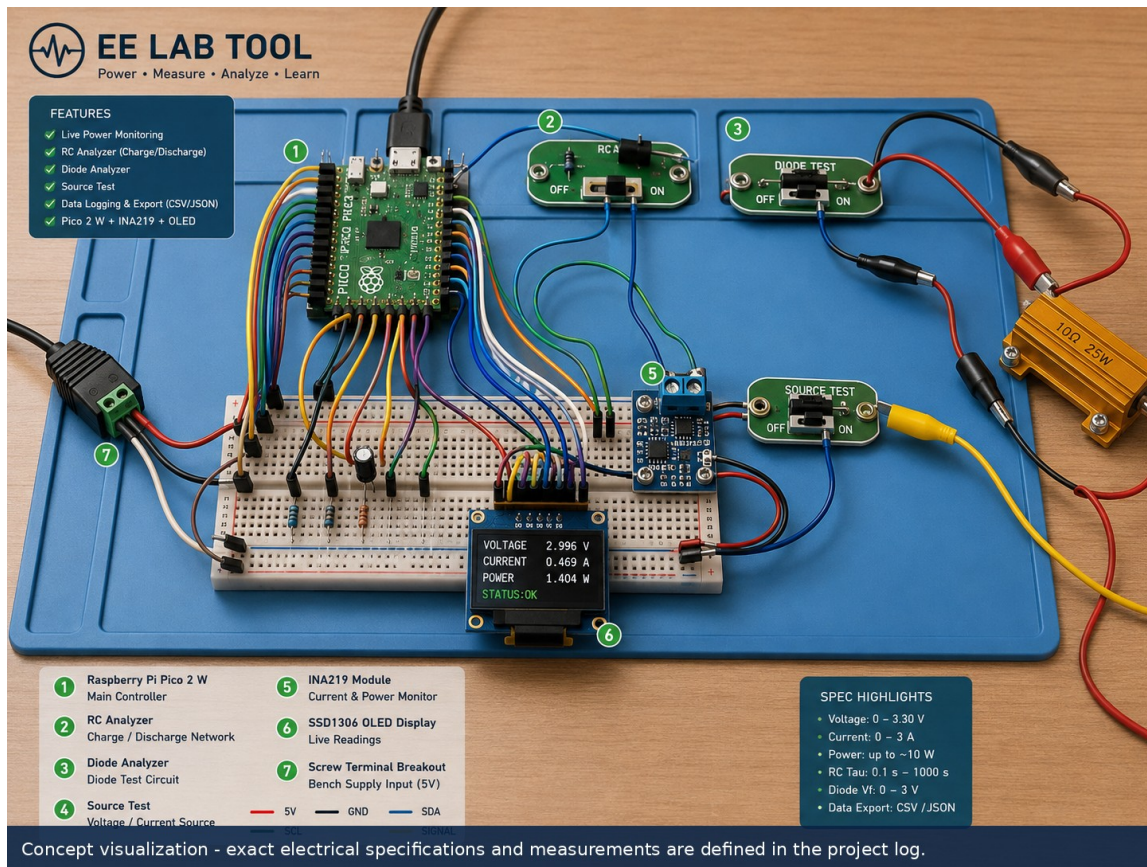
Annotated real-prototype overview.

EE LAB TOOL - VERIFIED SYSTEM ARCHITECTURE

Raspberry Pi Pico 2 W | MicroPython | SoftI2C 50 kHz | Wi-Fi dashboard



Exact 2D architecture and pin map.



Conceptual 3D/isometric visualization; exact specifications remain those in the text.

EE Lab Tool — Current State and GitHub Implementation Brief

Power/Energy Monitor • RC/Diode Analyzer • MCU Telemetry • Source Test • Fault Monitoring • Experiment Logging

Dashboard Overview (Reference Screenshots)

1) Completed Hardware

- Built multi-function EE test instrument on breadboard.
- Functions: power/energy monitor, RC analyzer, diode analyzer (single + I-V), source test, MCU temperature telemetry, and fault monitoring.
- MCU with on-board telemetry (internal pico die temperature).
- Measurement front-end with switchable circuits and measurement paths.
- Load testing uses 10 Ω 100 W aluminum power resistor.
- Power harness uses pluggable screw-terminal connector for clean, modular wiring.
- Status: Hardware is operational and integrated with firmware.

2) Implemented Software Modes

- Live Power Monitor:** voltage, current, power, charge (mAh), energy (Wh), min/max/avg, elapsed time.
- RC Analyzer:** measures time constant (tau) and capacitance.
- Diode Analyzer:** forward voltage measurement and classification (silicon vs. other).
- MCU Telemetry:** internal temperature sampling up to ~30 minutes.
- Source Test:** sweeps load points, fits Voc and source resistance.
- Fault Monitoring:** under/over-voltage and other limit checks.
- Experiment Logging:** stores results and fault events.
- Data Export:** CSV and JSON downloads for experiments and faults.

3) Verified Dashboard Features

- Live dashboard with real-time numeric tiles and trend charts.
- Panels for Power Monitor, RC Analyzer, Diode Analyzer, Source Test, MCU Telemetry, and Logs.
- Historical traces for voltage, current, and power since boot.
- RC charge/discharge curves and MCU temperature trend.
- Experiment log table with status and summary values.
- CSV/JSON export buttons for experiments and fault logs.

4) Key Measured Results (Examples)

- Live Power (Image A):** ~2.996 V, 0.469 A, 1.404 W.
- RC Analyzer (Image B):** tau = 1.079 s; capacitance = 1079 µF.
- Diode Analyzer (Image C):** classification: SILICON; Vf = 0.662 V at = 7.09 mA with 3.000 V test source.
- Source Test (Image D):** 3 points, fit → Voc = -0.003 V, source resistance = -6.395 ohm. (Negative R_s will be flagged for refinement/validation; not treated as final result.)
- MCU Telemetry (Image E):** internal die temperature = 41.25 °C ± 46.02 °C over ~30 minutes.
- Experiment Logs (Image F):** all test modes completed; fault monitor active.

6) Next GitHub Tasks

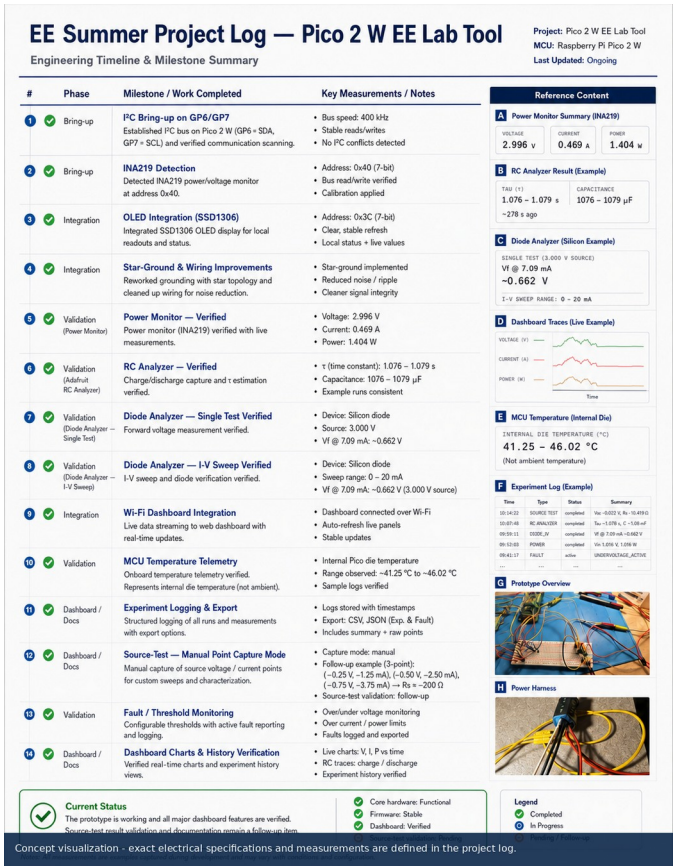
- Sanitize all screenshots (remove personal/system identifiers).
- Create /docs folder with organized documentation.
- Write README.md with architecture overview, features, and usage.
- Add hardware photo annotations (Image G) and power harness close-up (Image H).
- Include wiring diagrams, measurement block diagram, and BOM.
- Add dashboard screenshots with captions in /docs.
- Refine Source Test algorithm and validation logic.
- Prepare repository for GitHub Pages or repository publication.

Overall Prototype (System)

Power Harness Style (Close-up)

Concept visualization - exact electrical specifications and measurements are defined in the project log.

Conceptual one-page technical brief visual.



Conceptual project timeline visual.